

ML Centrality at FXT: Restructure to UrQMD+Detector+ $b < 14$, Centrality-Binning Fixes, and Phase-1 Dual-Evidential Modelling

Mathias Labonté

(working note accompanying the corresponding commit)

2026-05-29

1 Summary

This note documents three coupled changes pushed in this commit: (i) a scope restructure making **UrQMD + detector emulation + the full impact-parameter range $b < 14$ fm** the headline study; (ii) a fix to a centrality-binning off-by-one that affected both classical baselines and the ML labels, plus a refit of the Glauber baseline; and (iii) Phase 1 of the modelling work — two evidential heads (for b and N_{part}) replacing the previous regression+classification design, with centrality derived from predicted N_{part} . Full per-decision detail lives in `docs/restructure_urqmd_detector_b14.md` and `docs/modelling_plan.md`; this note gives the narrative and the rationale.

2 Data restructure (UrQMD + detector + $b < 14$)

What changed. The primary pipeline is now the UrQMD generator with the detector-emulated cache (`urqmd_padded_det.h5`: $0 < \eta_{\text{lab}} < 1.5$, p_T smearing, p_T threshold, logistic tracking efficiency), spanning the full $b < 14$ fm range at all four energies $\sqrt{s_{NN}} = 3.2, 3.5, 3.9, 4.5$ GeV (396,218 events). SMASH and the truth-level caches are retained but demoted to secondary comparisons. The shared cut/boost module `src/data/cuts.py` now sets `B_MAX_FM = 14`.

Why. The detector-emulated, full-range, transport-robust configuration is the one most representative of a real FXT measurement, so it is the right setting for the headline result rather than an idealized truth-level subset.

N_{part} **definition.** UrQMD’s *native* Glauber-header N_{part} is used directly (always ≥ 0). The earlier SMASH-style spectator recompute, $N_{\text{part}} = 2A - N_{\text{spec}}$ with $N_{\text{spec}} = \#\{\text{nucleon} \wedge n_{\text{coll}} = 0\}$, produced a negative peripheral tail in UrQMD: its `ncoll` counter does not inherit to decay-produced nucleons, so $\Delta \rightarrow N\pi$ daughters were mis-counted as spectators, driving $\sim 4.5\%$ of $b \in [11, 14]$ events to $N_{\text{part}} < 0$. That recompute existed only to make SMASH and UrQMD N_{part} comparable for joint training, which is now deprioritized; using the native value removes the pathology and lets us keep the full $b < 14$ range. We verified $\text{corr}(N_{\text{part}}, b) \approx -0.98$ in the cache, confirming it is a clean monotonic centrality variable.

Open caveat. The native N_{part} reaches a maximum of ~ 411 , and the ingest note computes its diagnostic with 2×208 (Pb), whereas Au has $A = 197$ ($2A = 394$). A value above 394 is impossible for Au+Au, so the nucleus / mass number behind UrQMD’s Glauber header must be confirmed before reporting N_{part} as a physical Au participant count. The headline target b is unaffected, and centrality binning is rank-based (hence robust to the absolute scale). N_{coll} remains deferred — it was not extracted at ingestion and the raw output is not locally available.

3 Classical baselines: binning fix and Glauber refit

Centrality-binning bug. The shared `assign_bins` (in `src/baselines/truth.py`, duplicated in `run_glauber.py`) computed the class index as `searchsorted(...)-1` followed by a clip. With seven percentile thresholds this merged the most-central stratum into bin 0, so the label “0–5%” actually held the top $\sim 10\%$ and every subsequent percentile label was shifted; a genuine most-central 5% class did not exist. Because the same helper produced the truth baseline, the Glauber baseline, *and* the ML classification labels, all three shared the identical (mis-labelled) binning — so relative comparisons were fair, but the absolute centrality labels were wrong. The fix uses `searchsorted(side="left")` without the -1 and sends events beyond the 80% edge to -1 (out of scope); the canonical helper is now imported in one place. Verified to reproduce the nominal 5/5/10/10/10/20/20% widths.

Glauber refit. The two-component NBD Glauber was refit per energy to the detector multiplicity with a low-multiplicity cut (`--fit-mult-min 11`, i.e. multiplicity > 10) that excludes the ultra-peripheral spike from the χ^2 window. The fitted-region residual is good at all energies (median $|\text{rel}| \approx 4\text{--}6\%$); a diagnostic was also corrected so the reported bulk residual respects the fit window (previously it was inflated by the excluded spike, e.g. 18% reported vs. $\sim 5\%$ true). Percentality $\langle b \rangle$ and $\langle N_{\text{part}} \rangle$ tables are now emitted and are consistent across energies, as expected for energy-independent geometry.

Bar to beat. On the fixed binning, the classical b -resolution (MAE per centrality class) is summarized in Table 1. The truth-tuned percentile baseline — the upper bound on any multiplicity-only classical method — is the meaningful target for the ML to beat.

method ($\sqrt{s_{NN}} = 3.2$)	0–5%	5–10%	10–20%	20–30%	30–40%	40–60%	60–80%
Truth-tuned	0.84	0.73	0.71	0.62	0.58	0.70	0.69
Glauber	0.86	0.75	0.77	0.72	0.76	0.92	0.70

Table 1: Classical-baseline b -MAE (fm) per centrality class at $\sqrt{s_{NN}} = 3.2$ GeV. Other energies are within a few percent. Truth-tuned $\approx 0.58\text{--}0.85$ fm is the bar.

4 Phase-1 modelling: dual evidential heads

Heads. The shared `OutputHeads` now carries *two* evidential Normal-Inverse-Gamma heads — one for b , one for N_{part} (μ, ν, α, β each) — and *no* classification head. Both targets are z -standardized on the train split and predicted in standardized space; the trainer inverts μ and scales σ back to physical units. The combined loss is $\mathcal{L} = \mathcal{L}_{\text{evid}}(b) + w \mathcal{L}_{\text{evid}}(N_{\text{part}})$.

Centrality from N_{part} , not b . Centrality is derived by ranking events on the chosen estimator and cutting the standard percentile edges: predicted N_{part} for the ML method, true N_{part} for the oracle, multiplicity for the classical baselines. N_{part} is the participant / volume measure that drives particle production and the volume fluctuations the analysis targets, so binning on it controls volume more tightly than b (at fixed b , N_{part} still has a Glauber spread) and matches the Glauber-centrality convention. This makes the cumulant chain internally consistent: classical bins on multiplicity, ML bins on predicted N_{part} , the oracle on true N_{part} .

The role of ϕ . The per-particle azimuth ϕ is kept as an input. It carries centrality information through the flow-vector magnitudes $|Q_n|$ (directed flow v_1 is large at FXT; elliptic v_2 is small and can be negative); only the absolute reaction-plane orientation Ψ_{RP} is the uniformly random nuisance.

We do *not* reconstruct the reaction plane — event-plane resolution is poor at FXT multiplicities. Instead a per-event random ϕ rotation augmentation (now applied in the canonical trainer) leaves all relative angles, and hence $|Q_n|$, invariant while randomizing Ψ_{RP} , forcing the network to learn the rotation-invariant flow structure rather than memorize an orientation it cannot generalize from. A $(\cos \phi, \sin \phi)$ input encoding is tracked as an ablation.

Pipeline. One canonical trainer (`train_cached.py`) covers all seven architectures (MLP, MLP-pool, DeepSets, Set Transformer, EFN, PFN, GNN), adds event-feature normalization, and drops the truth-label dependency for training (the truth baseline is now only a classical comparison). A two-epoch DeepSets smoke run validated the full path end-to-end (test b -MAE ≈ 0.53 fm, N_{part} -MAE ≈ 13), already in the regime of the classical bar; the proper apples-to-apples comparison will come from the evaluation step on fully-trained models.

5 Deferred / next

- Confirm the UrQMD collision system / mass number behind the native N_{part} header (Pb vs. Au); fall back to a b -only reported target if it proves Pb-based.
- $(\cos \phi, \sin \phi)$ encoding and ϕ -augmentation on/off ablations.
- Deep ensembles (5 seeds) for UQ robustness; cross-energy generalization (train on three energies, test on the fourth).
- Full training of all seven architectures, then rewire `validate_models.py` / `compute_cumulants.py` to the new prediction schema and the N_{part} -based centrality.
- N_{coll} as a third target (requires a cluster re-ingestion pass).